

Automated Network Request Management in ServiceNow

Solution Requirements

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Project Name	Automated Network Request Management in ServiceNow
Maximum Marks	4

1. Purpose

This document defines the functional and non-functional requirements for the **Automated Network Request Management in ServiceNow** project.

The solution is designed to automate the processing of network requests submitted through the ServiceNow Service Portal. It captures requester information, creates a corresponding Network Database record, sends the request for approval, processes the approval decision through workflow automation, updates the request status, and provides notifications to the relevant users.

The requirements described below are based on the functionality implemented in the project.

2. Functional Requirements

Requirements are grouped in the order they execute during a request's lifecycle, from intake through verification.

ID	Capability	What It Covers	Depends On
FR-01	Network Request Submission	The system shall provide a Network Request Catalog Item through the ServiceNow Service Portal so that users can submit network-related requests.	—
FR-02	Request Information Capture	The catalog item shall capture the required information provided by the requester, including requester details, mobile number, connection type, address, existing connection information where applicable, and other configured request details.	FR-01
FR-03	Mandatory Field Validation	The system shall ensure that all required fields are completed before the requester can submit the network request.	FR-02
FR-04	Conditional Field Handling	The Existing ID field shall be displayed only when the requester selects Existing as the connection type.	FR-02
FR-05	Automated Workflow Trigger	Once the Network Request is submitted, the configured ServiceNow Workflow/Flow Designer automation shall start automatically without requiring manual intervention.	FR-01 – FR-04

FR-06	Retrieve Request Details	The workflow shall retrieve the information submitted through the Network Request Catalog Item and use it for further processing.	FR-05
FR-07	Network Database Record Creation	The workflow shall automatically create a corresponding record in the Network Database using the submitted request information.	FR-06
FR-08	Request Data Mapping	The information entered by the requester shall be mapped correctly to the corresponding fields in the Network Database record.	FR-07
FR-09	Approval Request	After the Network Database record is created, the workflow shall generate an approval request for the configured approver.	FR-07
FR-10	Approval Decision	The approver shall be able to review the request and select an appropriate decision such as Approve or Reject .	FR-09
FR-11	Decision-Based Processing	The workflow shall use an If/Else decision to determine the next action based on the approval outcome.	FR-10
FR-12	Record Status Update	The Network Database record shall be updated automatically according to the approval decision.	FR-11
FR-13	Notification	The workflow shall send an email notification to the configured recipients after the request reaches the notification step.	FR-12
FR-14	Request Status Tracking	Users and authorised stakeholders shall be able to view the current status and approval information from the corresponding Network Database record.	FR-12
FR-15	Access Control	Access to the Network Database and related records shall be restricted to authorised ServiceNow users and roles.	FR-07
FR-16	Workflow Verification	The complete request lifecycle shall be tested to verify catalog submission, workflow execution, record creation, approval, decision handling, status update, and notification.	FR-01 – FR-15

3. Non-Functional Requirements

Quality Attribute	How the Solution Addresses It
Usability	The Network Request is provided through the ServiceNow Service Portal with a structured form, mandatory fields, and conditional field behaviour to make submission simple for the requester.
Automation	The request lifecycle is automated using ServiceNow Workflow/Flow Designer, reducing the need for manual processing after submission.
Security	Access to network request records is restricted through ServiceNow roles and access controls so that only authorised users can access or modify the records.
Reliability	The workflow is designed to consistently execute the configured sequence of record creation, approval, decision handling, status update, and notification.
Data Accuracy	Request information submitted through the catalog item is mapped to the Network Database record to maintain consistent and structured request data.
Performance	Automation reduces manual processing time and provides a faster request-handling process. No specific numerical response-time target was defined for this project.
Maintainability	The solution uses ServiceNow's standard catalog, database, workflow/flow, approval, notification, and access-control features, making the configuration easier to maintain.
Scalability	Additional network request fields, approval conditions, notifications, or workflow steps can be incorporated into the existing ServiceNow solution as requirements grow.

Availability	The solution operates within the ServiceNow platform and uses the platform's native services for catalog requests, workflow automation, records, approvals, and notifications.
Traceability	The request information, database record, approval outcome, and updated status are maintained within ServiceNow, allowing the request lifecycle to be tracked.

4. Out of Scope

The following functionality is outside the scope of the implemented project:

- Integration with external network-management systems.
- Real-time network device configuration.
- Machine-learning or AI-based network request prediction.
- External payment gateway integration.
- Mobile application development outside the ServiceNow platform.
- Custom performance or load-testing infrastructure.
- Advanced analytics beyond the ServiceNow functionality used in the project.